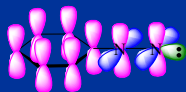
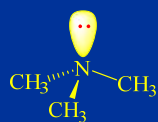


第十四章 含氮化合物

Nitrogen Compounds



1

CONTENT

- 1 硝基化合物
- 2 胺的结构和物理性质
- 3 胺的制备
- 4 胺的化学性质
- 5 季铵盐化合物
- 6 重氮与偶氮化合物

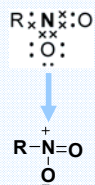
2

14.1 硝基化合物

硝基化合物的结构(了解)

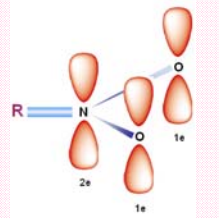
经典价键理论

N : $1s^2 2s^2 2p^3$

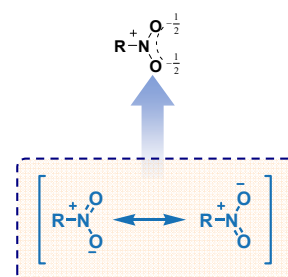


现代观点

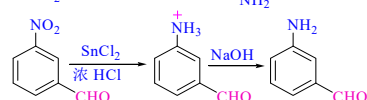
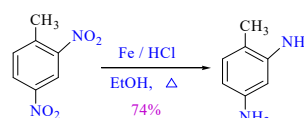
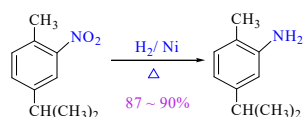
N : sp^2 杂化



Resonance structure



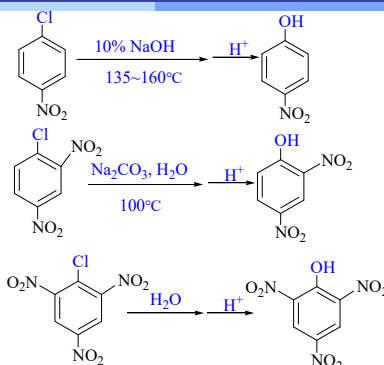
硝基化合物的还原反应



常用
Fe / HCl
Zn / HCl

5

硝基对氯苯水解影响



硝基对苯环上邻对位取代基的影响

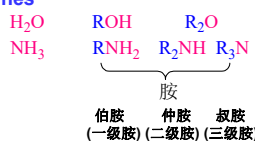
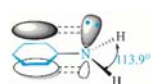
6

胺的结构和物理性质

• Structures of aliphatic amines



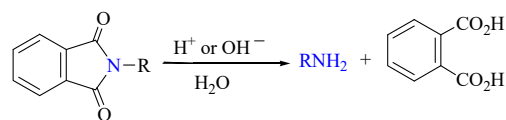
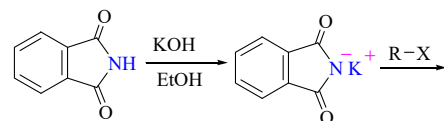
• Structures of aromatic amines



8

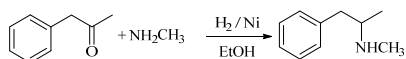
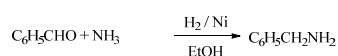
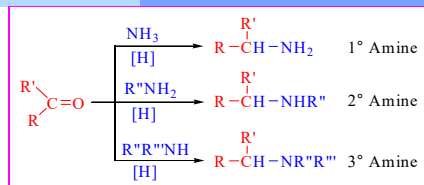
胺的制备

Gabriel 合成法



8

醛和酮的氨化还原

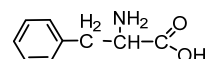
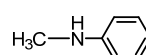


其它制备方式请自学

9

测试

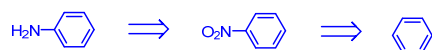
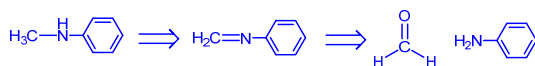
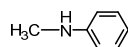
1) 以苯、甲苯以及不超过3个碳的有机试剂合成下列物质



10

测试

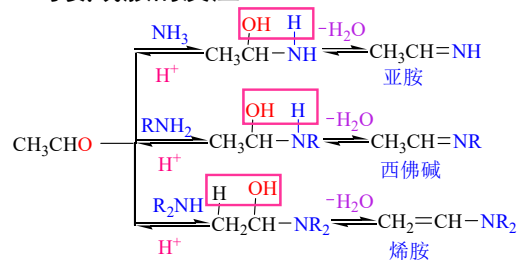
1) 以苯、甲苯以及不超过3个碳的有机试剂合成下列物质



11

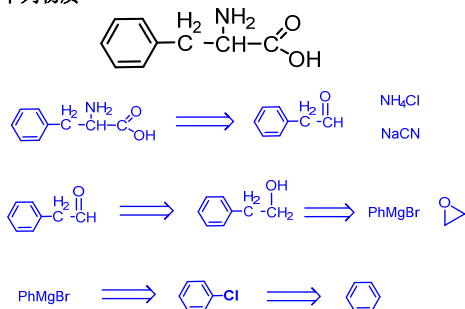
羰基与含氮亲核试剂的加成

1. 与氨或胺的反应 可逆反应, 分水操作

脂肪族Schiff 碱不稳定, 芳香族Schiff 碱稳定₁₂

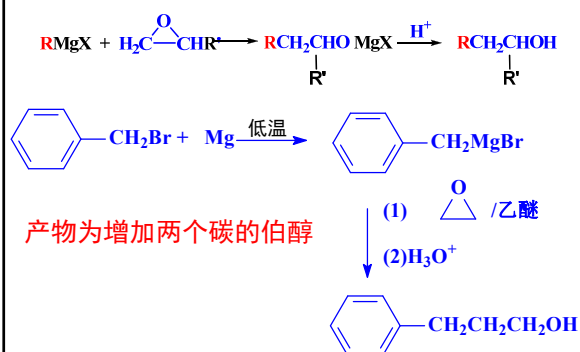
测试

1) 以苯、甲苯以及不超过3个碳的有机试剂合成下列物质

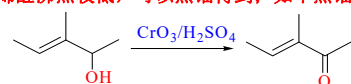
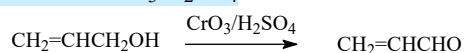


13

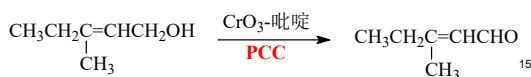
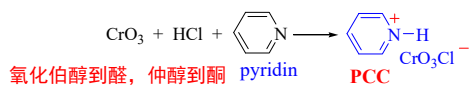
Example



Johns试剂: $\text{CrO}_3/\text{H}_2\text{SO}_4$



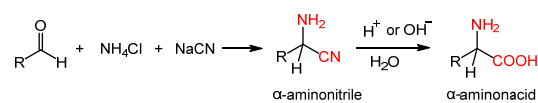
PCC (Pyridinium Chlorochromate)



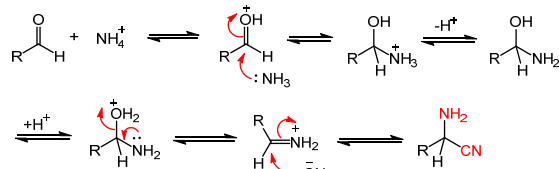
15

Strecker反应

羰基化合物与氯化铵、氰化钠反应, 形成 α -氨基腈, 经水解可以制备 α -氨基酸:



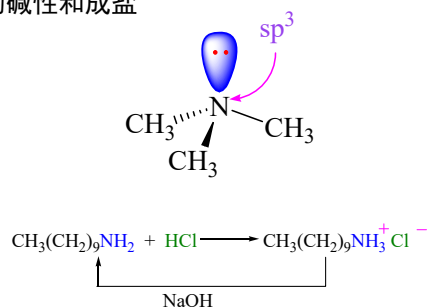
Mechanism:



16

胺的化学性质

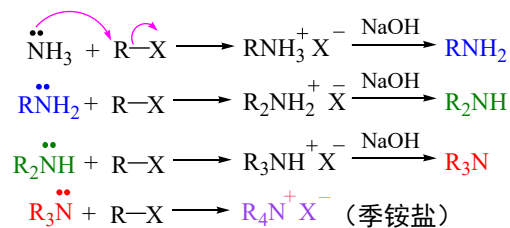
胺的碱性和成盐



17

胺的烃基化

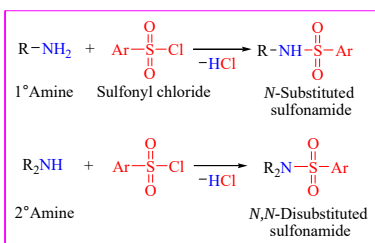
为什么不能用氯代烃直接制备伯仲叔胺?



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磺酰胺

Primary and secondary amines react with sulfonyl chlorides to form **sulfonamides**.

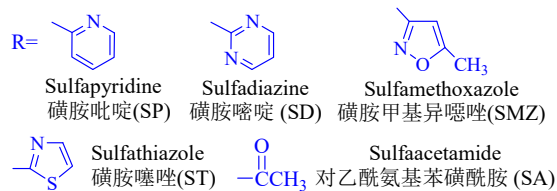
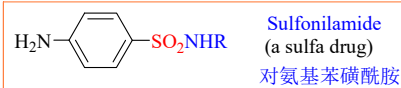


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磺酰胺药物

Sulfa Drugs

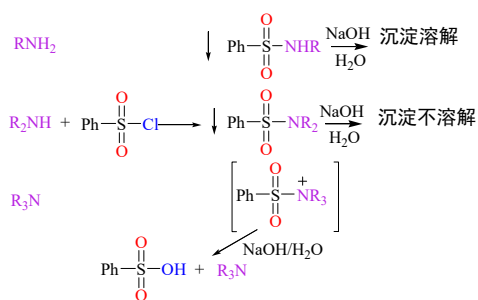
1932年发现
1935年临床应用



30

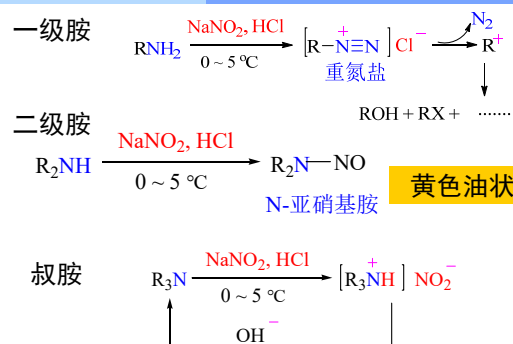
胺的分离鉴定 兴斯堡 (Hinsberg) 试验

胺的磺酰化：用于鉴别分离伯胺仲胺和叔胺



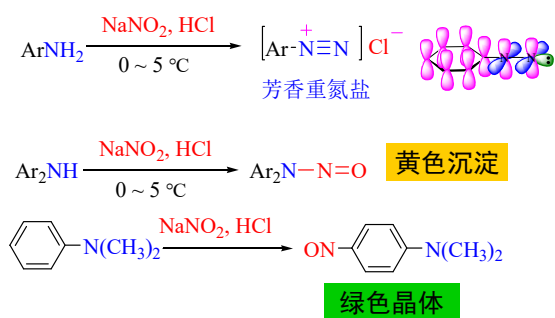
21

脂肪族胺与亚硝酸反应



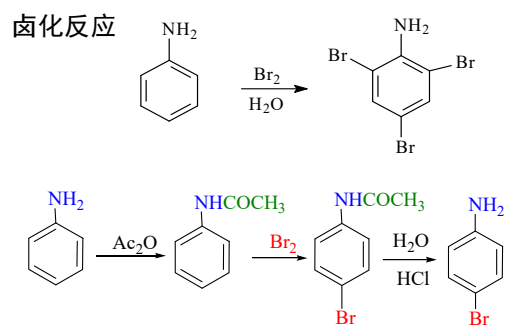
22

芳香胺与亚硝酸反应



23

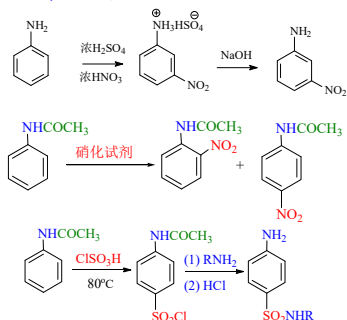
芳胺的亲电取代反应



24

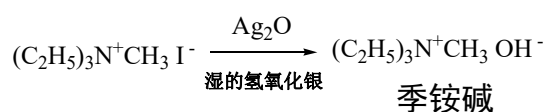
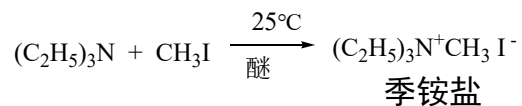
芳胺硝化、磺化反应

氨基在强酸作用下，由活化基团变成了钝化基团，因此最好将氨基保护后，再进行硝化、磺化反应



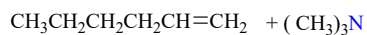
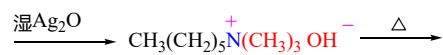
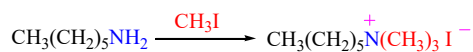
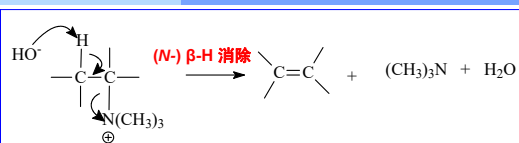
25

季铵盐化合物



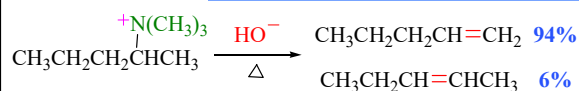
26

季铵碱受热分解反应



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霍夫曼 (Hofmann) 消除

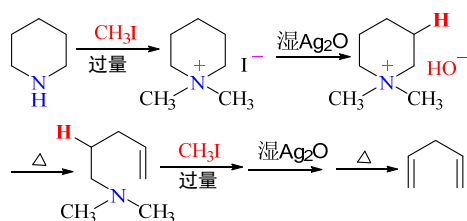


霍夫曼 (Hofmann) 消除

当季铵碱分子中有两个以上不同的氢原子可被消除时，反应主要从含氢较多的碳原子上消去氢原子，即主要生成双键碳原子上烷基取代较少的烯烃

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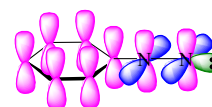
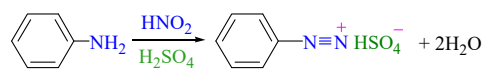
应用：推测胺的结构



29

重氮与偶氮化合物

重氮盐的制备——重氮化反应

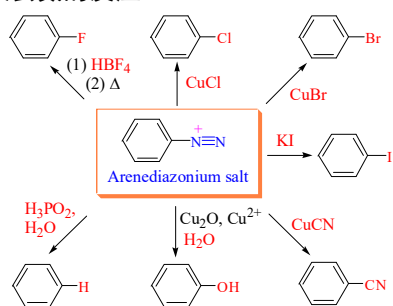


- 重氮盐是无色晶体，在干燥条件下不稳定，爆炸性强
- 重氮盐一般不溶于有机溶剂，可溶于水，其水溶液呈中性，在水溶液中发生离子化，溶液具有导电性
- 重氮化反应要在酸性条件下进行，酸的用量一般为1.5倍

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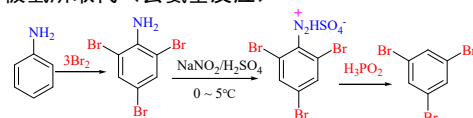
芳香重氮盐在合成上的应用

(1) 失去氮的反应

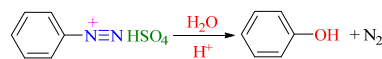


31

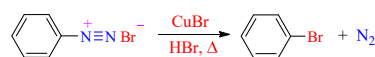
被氢所取代（去氨基反应）



被羟基所取代



被卤素取代

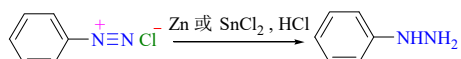


32

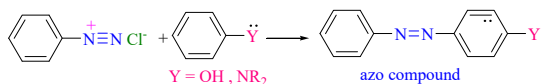
偶氮化合物制备

(2) 保留氮的反应

• 还原反应



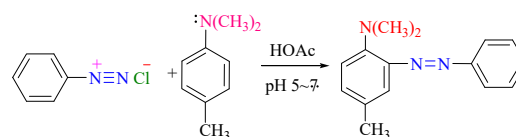
• 偶联反应



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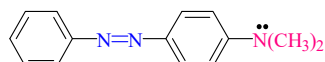
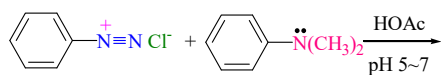
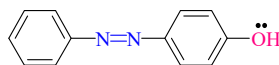
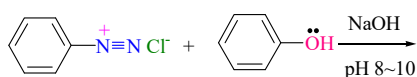
偶联反应

在微酸性中性或微碱性溶液中，重氮盐正离子作为亲电试剂可与连有强供电基的芳香族化合物，如酚芳胺等发生亲电取代反应生成偶氮化合物



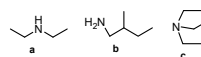
34

Example



作业

1. 用简单的化学方法鉴别下列各组化合物

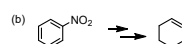
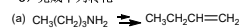


2. 排序题

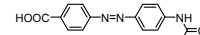
按碱性由强到弱排序



3. 完成下列转化



4. 以苯、甲苯、C2以及以下有机物合成下列物质



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